DAN MILL TRAINING

© Dan Mill Training

RAID Types Cheat Sheet

RAID 0 (Striping)

What it does: Splits data across multiple drives for speed

Minimum drives: 2 **Capacity**: 100% of all drives combined **Fault tolerance**: None - if any drive fails, all data is lost

Performance: Excellent read/write speeds **Best for**: High-performance applications, video editing, gaming **Use when**: Speed is priority and data is backed up elsewhere

---

RAID 1 (Mirroring)

What it does: Creates exact copies of data on multiple drives

Minimum drives: 2 **Capacity**: 50% of total drive space **Fault tolerance**: Can lose 1 drive

Performance: Good read speeds, normal write speeds **Best for**: Critical data storage, operating system drives **Use when**: Data protection is the top priority

---

RAID 5 (Striping with Parity)

What it does: Distributes data and parity information across all drives

Minimum drives: 3 **Capacity**: Total capacity minus 1 drive (e.g., 3x1TB = 2TB usable) **Fault tolerance**: Can lose 1 drive

Performance: Good read speeds, slower write speeds due to parity calculations **Best for**: File servers, general business storage **Use when**: Need balance of performance, capacity, and protection

---

RAID 6 (Striping with Double Parity)

What it does: Like RAID 5 but with two parity drives worth of protection

Minimum drives: 4 **Capacity**: Total capacity minus 2 drives (e.g., 4x1TB = 2TB usable) **Fault tolerance**: Can lose 2 drives

Performance: Good read speeds, slower write speeds than RAID 5 **Best for**: Large storage arrays, mission-critical data **Use when**: Need protection against multiple drive failures

---

RAID 10 (1+0) - Mirrored Stripe

What it does: Combines RAID 1 mirroring with RAID 0 striping

Minimum drives: 4 (even numbers only) **Capacity**: 50% of total drive space **Fault tolerance**: Can lose 1 drive from each mirror pair

Performance: Excellent read/write speeds **Best for**: Database servers, high-performance applications **Use when**: Need both speed and redundancy with good budget

---

RAID 01 (0+1) - Striped Mirror

What it does: Creates stripes first, then mirrors the stripe sets

Minimum drives: 4 (even numbers only) **Capacity**: 50% of total drive space **Fault tolerance**: Can lose all drives in one stripe set

Performance: Good read/write speeds **Best for**: Less common, RAID 10 is generally preferred **Use when**: Rarely recommended over RAID 10

Quick Comparison

RAID Level	Min Drives	Usable Space	Speed	Fault Tolerance	Best For
RAID 0	2	100%	Excellent	None	Performance
RAID 1	2	50%	Good	1 drive	Critical data
RAID 5	3	n-1 drives	Good read	1 drive	General business
RAID 6	4	n-2 drives	Good read	2 drives	Large arrays
RAID 10	4	50%	Excellent	1 per mirror	High-performance

Hardware vs Software RAID

Hardware RAID:

- Dedicated RAID controller card
- Better performance, especially for RAID 5/6
- Independent of operating system
- More expensive
- Hot-swap capability

Software RAID:

- Built into operating system
- Uses system CPU for calculations

- Less expensive
- Easier to move between systems
- May impact system performance

Remember: RAID is NOT a backup solution - always maintain separate backups!